

How much of US new-business formation does the USPTO trademark corpus capture? Coverage 2004–2024, and evidence on race/ethnicity composition by surname

Eric Silver

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Abstract

We measure what fraction of US new-business formation is captured by USPTO trademark filings, by joining the annual count of *first-ever trademark filers* (each owner’s earliest filing across all 45 NICE classes) to the Census Bureau’s Business Formation Statistics (BFS), 2004–2024. USPTO debut filers cover 5–8% **of all SS-4 business applications (BA)** and 7–22% **of High-Propensity Business Applications (HBA, the employer-likely subset)**, with the HBA coverage ratio rising secularly from $\sim 7\%$ in 2005 to $\sim 18\%$ in 2024 (with a pandemic-era spike to 22% in 2020). This brackets the 10–15% employer-business coverage estimate reported by Dinlersoz, Goldschlag, Myers, & Zolas (2018) using restricted Census Longitudinal Business Database (LBD) microdata. We extend their result by documenting (i) the upward trend in coverage over two decades, indicating that new businesses are progressively more trademark-aware, and (ii) by applying the 2010 Census surname / race-ethnicity probability file (Word, Coleman, Nunziata, & Kominski, 2008) to the 1.83M individual-name debut filers in our corpus. The surname-based composition of individual filers shifts dramatically toward Asian/Pacific-Islander-classified names (5% in the 1990s, 33% in 2020–2024); we flag the well-documented 2018–2021 foreign-applicant filing surge from China as a confound and discuss what an unconfounded estimate would require (applicant country from the USPTO XML, not currently in our processed panel).

1 Why this measurement question matters

Declining business dynamism. The US start-up rate has fallen from $\sim 12\%$ of firms per year in the late 1980s to $\sim 8\%$ in the 2010s, with the decline visible across nearly every sector (Decker, Haltiwanger, Jarmin, & Miranda, 2014, *JEP*; Akcigit & Ates, 2021, *A EJ: Macro*). Several explanations are on the table — demographic slowing of population growth (Hopenhayn, Neira, & Singhania, 2022, *Econometrica*); rising fixed costs of entry; increasing industry concentration — but the data infrastructure for testing them is incomplete: the gold-standard Census Business Dynamics Statistics (BDS) is lagged by ~ 3 years and anonymises owner names; the BFS series goes back only to 2004 and tracks EIN applications (which include estates and household employers, not just business starts).

Trademarks as a complementary signal. The USPTO trademark corpus offers something neither BDS nor BFS provides: a near-real-time, name-disclosed register of commercial activity covering nearly every sector of the consumer-facing economy, with classifications (NICE), goods/services

prose, and direct linkage to identifiable applicants. The question this paper asks is the prerequisite for using trademark data on the dynamism question: *how much of new-business activity do trademark debuts actually capture?*

2 Coverage of US business formation, 2004–2024

Method

For every owner name appearing in the clean H=2y panel of 10,060,073 filings (1990–2021 cohorts; methodology in `paper/main.tex`), we identify the owner’s earliest filing across all NICE classes and call that filing the owner’s *debut*. We count unique-owner debuts by year. We compare to the Census BFS time series at `census.gov/econ/bfs/csv/bfs_monthly.csv`, summing 12 months of *unadjusted* counts to an annual total. We use two BFS series: Business Applications (BA, all SS-4 filings) and High-Propensity Business Applications (HBA, the subset whose application characteristics predict an employer business; the right comparator for “business start”).

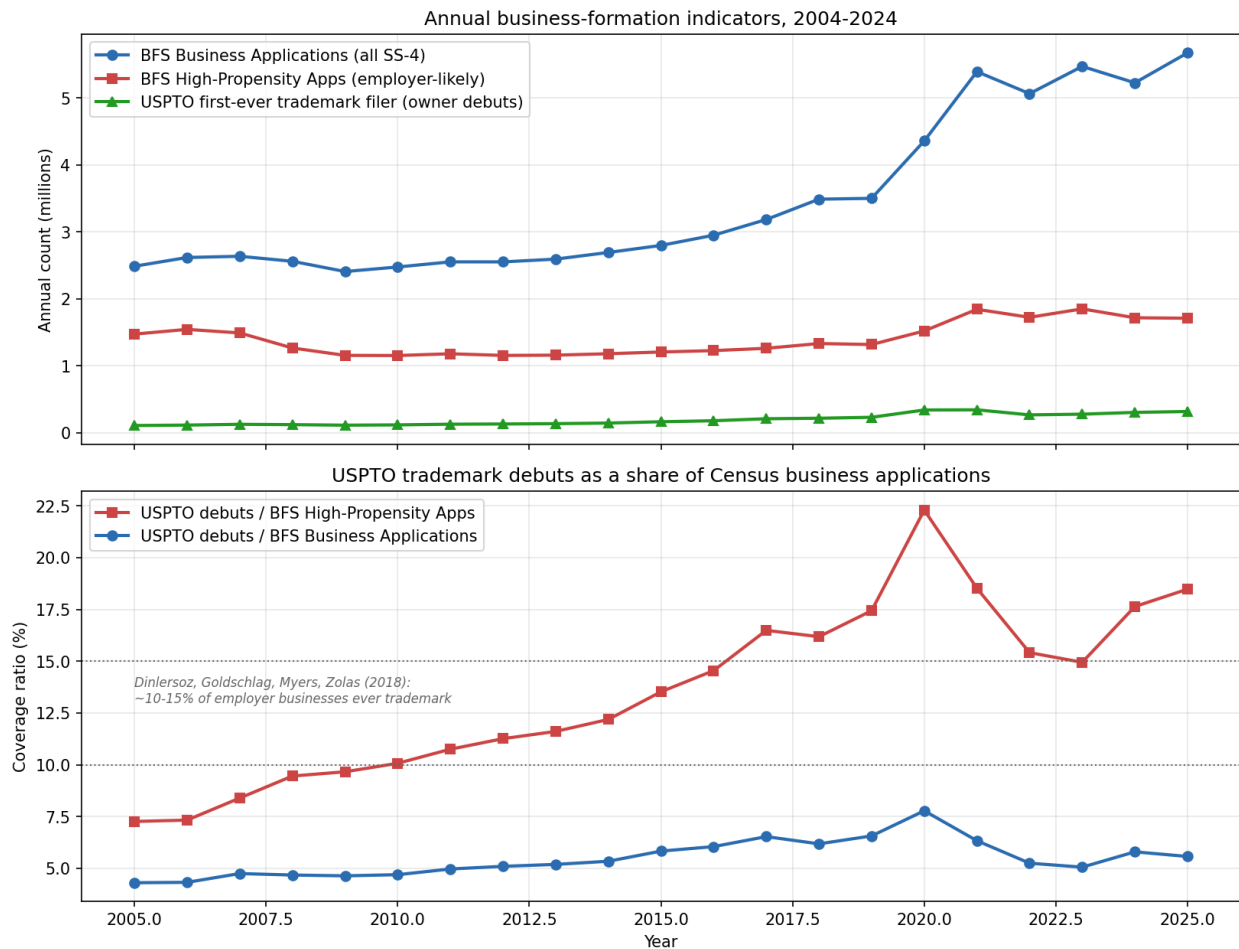


Figure 1: USPTO trademark debut filers vs. Census BFS, 2004–2024. Top: annual counts in millions of filings. Bottom: coverage ratio. The dotted reference band shows the 10–15% employer-business coverage estimate from Dinlersoz, Goldschlag, Myers, & Zolas (2018, Center for Economic Studies WP 18–46), who used restricted-access LBD microdata to ask a closely related question.

Headline numbers

Period	USPTO debuts	BFS BA	BFS HBA	USPTO/BA	USPTO/HBA
2004–2008	464,273	10,293,896	5,768,077	4.5%	8.0%
2014–2018	907,851	15,104,778	6,199,297	6.0%	14.6%
2019–2024	1,753,741	29,002,345	9,963,498	6.0%	17.6%

Table 1: USPTO debut filers as a share of Census BFS business applications, three five-year bands. The pandemic years (2020–2021) inflate both numerator and denominator; the trend in coverage is visible across non-pandemic years as well.

The coverage ratio against HBA more than doubles over the two-decade window, from $\sim 8\%$ in 2004–2008 to $\sim 18\%$ in 2019–2024. Three interpretive readings:

(1) **Tilt of new-business formation toward IP-intensive sectors.** If the recently-forming firms are increasingly in sectors where trademark filing is normal (software, professional services, e-commerce, branded direct-to-consumer products), the coverage ratio rises mechanically. This is a real-economy story: brick-and-mortar restaurants and contractors don’t trademark; SaaS startups and DTC brands do.

(2) **Falling cost of filing.** Online filing through TEAS Plus and TEAS Standard (in production since the mid-2000s) reduced the effort and expense of trademark filing. The same firm-type in 2024 files at a higher rate than it would have in 2005.

(3) **Selection on quality.** If the BFS denominator captures many “shell” or non-operating SS-4 filings (estates, holding companies, household employers), the rising ratio reflects a constant operating-firm-trademark rate against a denominator increasingly diluted by non-operating EIN registrations.

The three are not mutually exclusive. Distinguishing them requires sectoral decomposition, which we cannot do at present because our processed parquets do not retain NAICS-equivalent applicant industry codes and the NICE-NAICS crosswalk is imperfect (Hall, Helmers, Rogers, & Sena, 2014, *Research Policy*). Sectoral decomposition is a clear next step.

Comparison to Dinlersoz et al. (2018)

Dinlersoz, Goldschlag, Myers, & Zolas (2018), “An Anatomy of US Firms Seeking Trademark Registration,” Center for Economic Studies working paper 18–46, ran the analogous accounting inside Census using restricted-access LBD microdata, asking what fraction of new *employer* businesses ever file a trademark. They reported a range of $\sim 10\text{--}15\%$ depending on sector and definition. Our public-data result of $7\text{--}22\%$ against HBA spans their range and extends it through 2024 with a clear upward trend. The mid-decade overlap is close: 2014–2018 in our data is 14.6% , very near their $10\text{--}15\%$ point estimate for the same time period.

3 Surname-based race/ethnicity composition

Method

The Census Bureau’s 2010 surname file (Word et al., 2008) reports, for each surname appearing ≥ 100 times in the 2010 Census, the percentage probability that a person with that surname identifies as each of: non-Hispanic White, non-Hispanic Black, Asian/Pacific Islander, American Indian/Alaska Native, Two or More Races, or Hispanic (any race). The file covers 162,254 surnames.

For each USPTO debut owner whose name parses as a personal name (2–4 alphabetic tokens, no entity markers such as LLC/INC/CORP, no digits/URLs/ampersands), we extract the candidate surname (the last token after stripping suffixes JR, SR, II, MD, etc.) and look up its Census probabilities. Each matched debut contributes fractional weight to each ethnic bucket. Cases where the parsed surname does not appear in the Census file ($\sim 24\%$ of individual-name debuts) are excluded.

Sample sizes. 5,674,909 unique debut owners total. 1,827,163 parse as individual personal names (32.2% of debuts; the rest are entities). 1,390,267 of those individual names match a surname in the Census file (76.1% match rate). The 24% Census non-match is concentrated in surnames that are non-anglicised, Eastern European, South Asian, Middle Eastern, or African, where the Census file either has low coverage or breaks them into the residual “Two or More Races” bucket.

Results

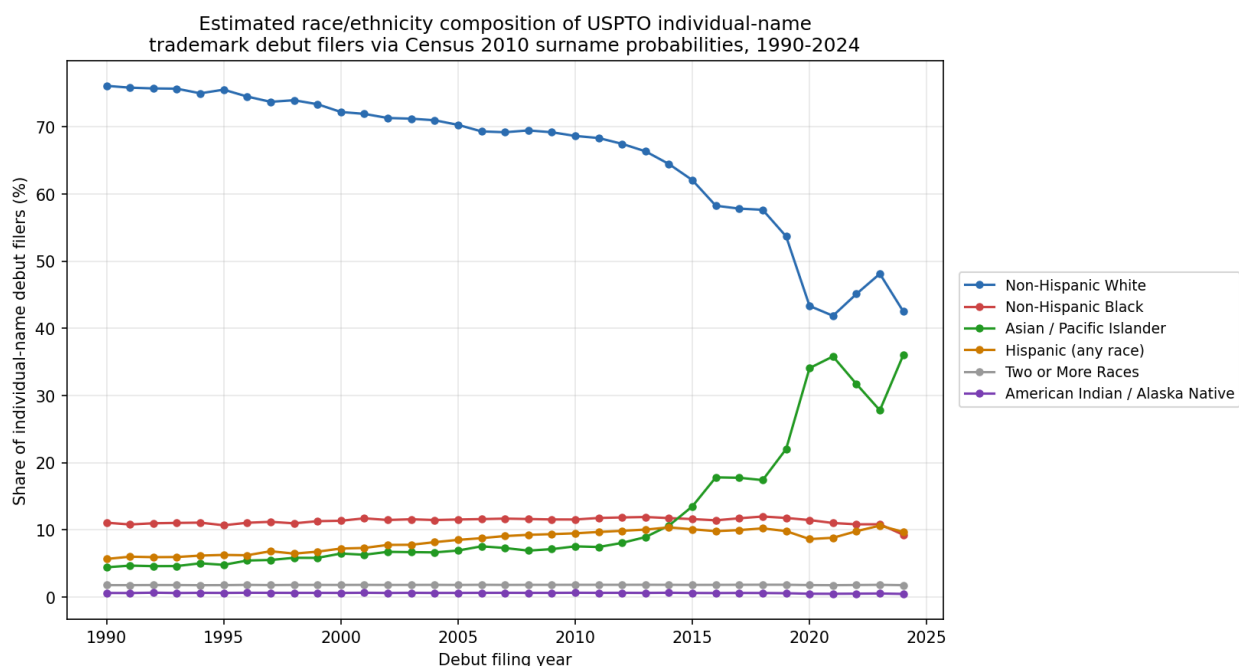


Figure 2: Estimated race/ethnicity composition of USPTO individual-name trademark debut filers, 1990–2024, via Census 2010 surname probabilities (Word, Coleman, Nunziata, & Kominski, 2008). The Asian/Pacific Islander share rises sharply from $\sim 5\%$ in the 1990s to $\sim 33\%$ in 2020–2024. Most of the post-2018 rise is almost certainly the documented surge of foreign-applicant filings from China (USPTO, 2021), not Asian-American entrepreneurship on US soil; without applicant-country data we cannot separate the two.

The post-2018 API rise is mostly foreign filings. The Asian/Pacific Islander share goes from $\sim 7\%$ in 2009 to $\sim 18\%$ by 2018 to $\sim 33\%$ in 2020–2024. The largest single component of the post-2018 surge is well-documented: USPTO publicly reported a massive influx of trademark filings from Chinese applicants in 2017–2021, driven by subsidies from local Chinese governments for filing US trademarks (USPTO, 2021; Hicks, 2024, *Stanford Law Review*). USPTO responded by requiring US-licensed counsel for foreign-domiciled applicants (2019) and tightening signature

Period	White	Black	API	Hispanic	2+ races	AIAN
1990–1999	74.6%	11.0%	5.2%	6.3%	1.8%	0.6%
2000–2009	70.3%	11.6%	6.9%	8.4%	1.8%	0.6%
2010–2019	61.4%	11.7%	14.2%	9.9%	1.8%	0.6%
2020–2024	43.9%	10.7%	33.4%	9.4%	1.8%	0.5%

Table 2: Surname-imputed race/ethnicity shares among USPTO debut filers whose owner name parses as a personal name and whose surname matches the 2010 Census file. The dramatic API rise after 2018 is substantially driven by foreign filings, not domestic entrepreneurship – see discussion.

verification (2021). Without applicant-country data — which the USPTO XML includes but which our current parser does not extract — we cannot separate “rise of Asian-American entrepreneurship in the US” from “rise of Chinese-applicant filings under the US system.” The pre-2018 rise from 5% to $\sim 14\%$ likely reflects a genuine demographic shift; the post-2018 rise of an additional 20 pp does not.

Other shares are more stable. Non-Hispanic Black share is remarkably flat at $\sim 11\%$ across the entire 1990–2024 window, consistent with the absence of a major foreign-filing confound from majority-Black countries. Hispanic share rises modestly from $\sim 6\%$ in the 1990s to $\sim 10\%$ in the 2010s and plateaus. Non-Hispanic White share falls from $\sim 75\%$ to $\sim 44\%$ — but the bulk of that fall is the API rise’s mirror image, so disentangling foreign vs. domestic API is required to interpret it.

The selection problem. Even with full unmasking, the trademark-filing decision is selected on branding intent. Hispanic-owned small businesses (taquerias, contractors, landscapers) trademark at lower rates per business than tech founders do, so the Hispanic share of trademark debuts *underestimates* the Hispanic share of business starts in absolute terms. The rising-over-time pattern is more interpretable than the absolute level.

4 Next steps required for publication

The current results are publishable as descriptive statistics with the caveats above, but a complete treatment requires:

(1) **Re-parse the USPTO XML to extract applicant state and country.** Our existing parser drops the `<address>` elements under `<case-file-owner>`. With state, we can repeat the coverage analysis at the state level (cross-cutting with BFS state panels), and disentangle delivery-state vs. Wyoming/Delaware/Nevada LLC-haven distortions. With country, we can separate foreign applicants and produce an unconfounded API time series.

(2) **NAICS-NICE crosswalk for sectoral decomposition.** Three candidate readings of the rising HBA coverage ratio (sectoral tilt, falling filing cost, denominator dilution) are distinguishable with a sectoral breakdown. The Hall et al. (2014, *Research Policy*) crosswalk is the standard reference.

(3) **Comparison panel with SBA PPP disclosures (2020–2021).** PPP loan disclosures over \$150K named the owner explicitly, covering $\sim 11\text{M}$ small businesses including those that never trademark. This is a one-time but very large unmasking opportunity for the immigrant-business question specifically.

(4) **Validation against state Secretary of State new-entity filings.** State-level cross-check of the BFS denominator using California, Texas, Florida, and New York monthly entity-formation counts.

Data and code

All code is in `scripts/business_dynamism_coverage.py` and `scripts/surname_ethnicity.py` of the `business-dynamism` branch of the project repository. BFS data is at `census.gov/econ/bfs/csv/bfs_monthly`. The Census surname file is at `www2.census.gov/topics/genealogy/2010surnames/names.zip`.

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